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Batch: 15

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Experiment-36

Algorithm

1. Start.
2. Declare `choice`.
3. Display the menu:
 - Count characters.
 - Print each character on a new line.
 - Count vowels and consonants.
 - Exit.
4. Read the user's choice.
5. Use `switch` to perform the selected operation.
6. For **Choice 1**, call `countCharacters()`:
 - Read the string.
 - Start `i = 0`.
 - Traverse the string until `'\0'` or `'\n'`.
 - Increment `i` for every character.
 - Display the character count.
7. For **Choice 2**, call `printCharacters()`:
 - Read the string.
 - Traverse each character.
 - Print each character on a separate line.
8. For **Choice 3**, call `countVowelsConsonants()`:
 - Read the string.
 - Check every character.
 - If it is `a, e, i, o, u` in uppercase or lowercase, increment the vowel count.
 - If it is another alphabetic character, increment the consonant count.
 - Display both counts.
9. For **Choice 4**, exit the program.
10. For any other choice, display `"Invalid choice!"`.
11. Repeat the menu until Choice 4 is selected.
12. Stop.

Test Case Table

TC	Choice	Input String	Expected Output
TC1	1	Hello	Number of characters = 5
TC2	1	Hello World	Number of characters = 11
TC3	2	ABC	A, B, C each on a new line
TC4	3	Hello	Vowels = 2, Consonants = 3
TC5	3	AEIOU	Vowels = 5, Consonants = 0
TC6	3	123 abc!	Vowels = 1, Consonants = 2
TC7	4	—	Exiting program...
TC8	5	—	Invalid choice!

C Program

```
#include <stdio.h>
```

```
void countCharacters();  
void printCharacters();  
void countVowelsConsonants();
```

```
int main()  
{  
    int choice;  
  
    do  
    {  
        printf("\n--- STRING OPERATIONS ---\n");  
        printf("1. Count characters\n");  
        printf("2. Print each character on a new line\n");  
        printf("3. Count vowels and consonants\n");  
        printf("4. Exit\n");  
  
        printf("Enter your choice: ");  
        scanf("%d", &choice);  
        getchar();  
  
        switch(choice)  
        {  
            case 1:  
                countCharacters();  
                break;  
  
            case 2:  
                printCharacters();  
                break;
```

```

        case 3:
            countVowelsConsonants();
            break;

        case 4:
            printf("Exiting program...\n");
            break;

        default:
            printf("Invalid choice!\n");
    }

    } while(choice != 4);

return 0;
}

void countCharacters()
{
    char str[100];
    int i = 0;

    printf("Enter a string: ");
    fgets(str, 100, stdin);

    while(str[i] != '\0' && str[i] != '\n')
    {
        i++;
    }

    printf("Number of characters = %d\n", i);
}

void printCharacters()
{
    char str[100];
    int i = 0;

    printf("Enter a string: ");
    fgets(str, 100, stdin);

    printf("Characters are:\n");

    while(str[i] != '\0' && str[i] != '\n')
    {
        printf("%c\n", str[i]);
        i++;
    }
}

```

```

void countVowelsConsonants()
{
    char str[100];
    int i = 0, vowels = 0, consonants = 0;

    printf("Enter a string: ");
    fgets(str, 100, stdin);

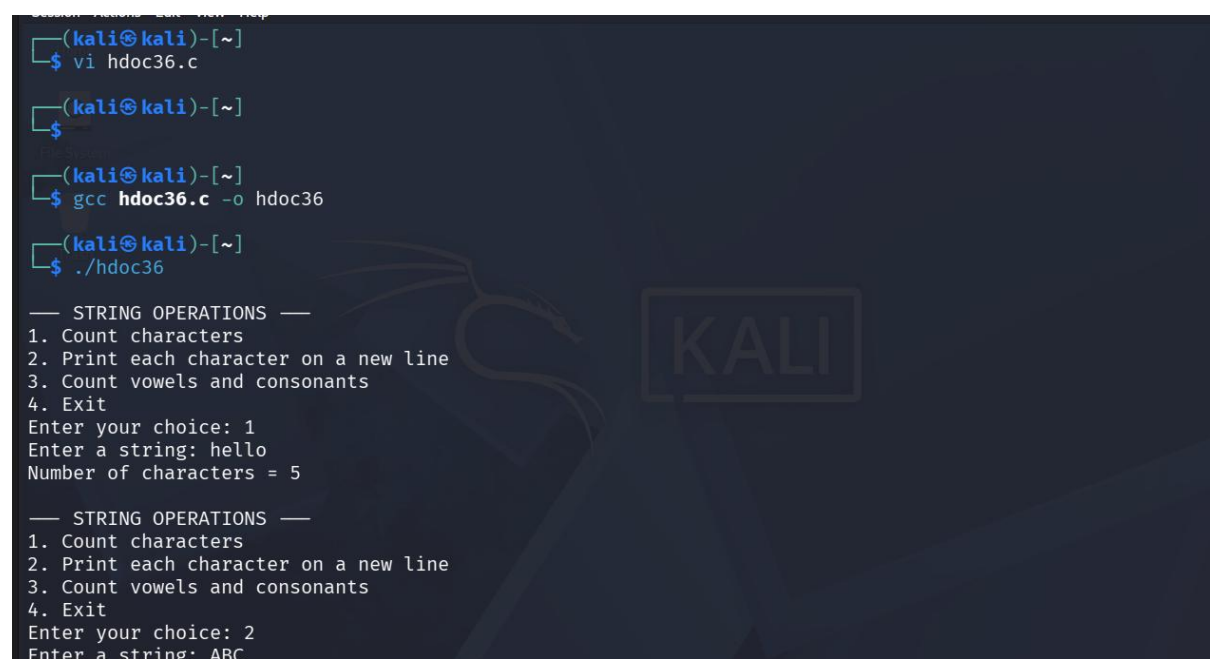
    while(str[i] != '\0' && str[i] != '\n')
    {
        if(str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
           str[i] == 'o' || str[i] == 'u' || str[i] == 'A' ||
           str[i] == 'E' || str[i] == 'I' || str[i] == 'O' ||
           str[i] == 'U')
        {
            vowels++;
        }
        else if((str[i] >= 'a' && str[i] <= 'z') ||
                (str[i] >= 'A' && str[i] <= 'Z'))
        {
            consonants++;
        }

        i++;
    }

    printf("Number of vowels = %d\n", vowels);
    printf("Number of consonants = %d\n", consonants);
}

```

Compilation And Execution



```

(kali㉿kali)-[~]
└─$ vi hdoc36.c

(kali㉿kali)-[~]
└─$

(kali㉿kali)-[~]
└─$ gcc hdoc36.c -o hdoc36

(kali㉿kali)-[~]
└─$ ./hdoc36

— STRING OPERATIONS —
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Enter a string: hello
Number of characters = 5

— STRING OPERATIONS —
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 2
Enter a string: ABC

```

Enter a string: ABC

Characters are:

A

B

C

— STRING OPERATIONS —

1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 3

Enter a string: 123 abc!

Number of vowels = 1

Number of consonants = 2

— STRING OPERATIONS —

1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 4

Exiting program...